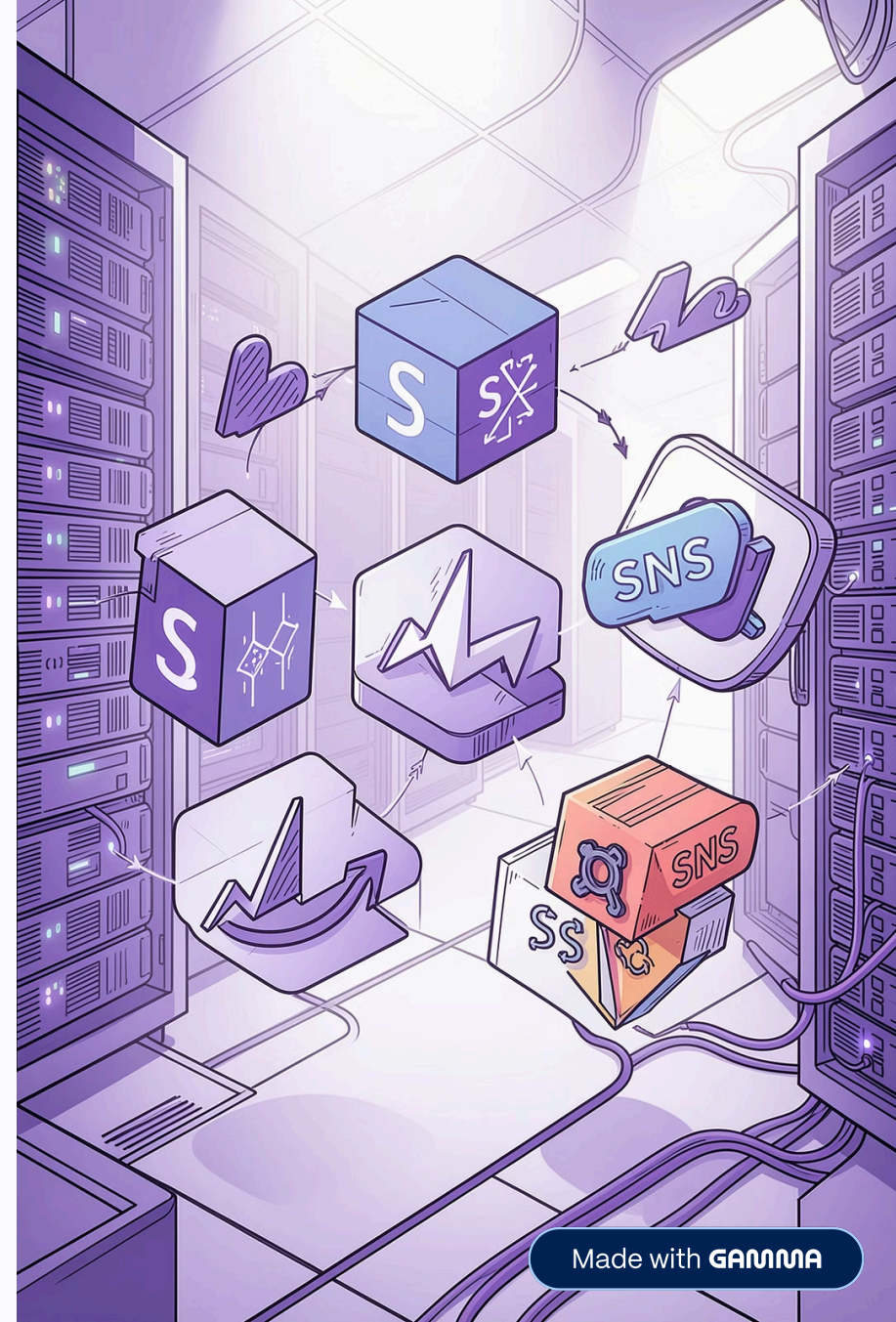


Serverless File Notifications: S3 → Lambda → SNS

Concise overview of a lightweight, event-driven pipeline that sends real-time email notifications when files are uploaded to Amazon S3. Designed for reliability, low operational overhead, and near-instant delivery.

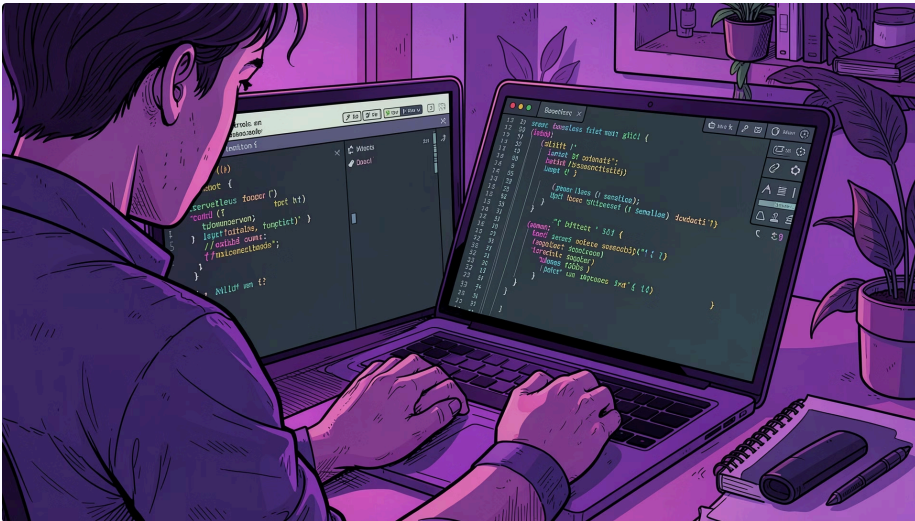


Architecture At a Glance



The system is fully event-driven. S3 raises an ObjectCreated event, Lambda handles transformation and enrichment, and SNS performs reliable fan-out to email endpoints.

Event Flow & Lambda Responsibilities



1. Receive S3 Event

Lambda is invoked with the S3 ObjectCreated event (JSON). Includes bucket name, object key, and event time.



2. Extract & Enrich

Parse object key, call S3 HEAD if needed to confirm size/metadata, compute human-friendly timestamp and file summary.



3. Publish to SNS

Format a structured message (JSON + plain-text subject/body) and publish to the SNS topic for delivery to email subscribers.

Operational Considerations

Idempotency

Design Lambda to tolerate duplicates — dedupe by object key + ETag or versionId when necessary.

Error Handling

On SNS publish failures, retry with exponential backoff. Send failed events to DLQ (SQS) for later reprocessing.

Security

Least-privilege IAM: Lambda needs s3:GetObject or s3:HeadObject and sns:Publish to the topic. Encrypt SNS topics if required.

Monitoring

Track Lambda failures, throttles, and SNS delivery metrics (DeliveryAttempts, NumberOfMessagesPublished). Alert on elevated error rates.

Deployment Checklist & Next Steps



IAM Roles

Create a minimal role for Lambda with `s3:HeadObject` and `sns:Publish`.



Infrastructure as Code

Deploy S3 notification, Lambda, and SNS via CloudFormation / SAM / CDK for repeatability.

Suggested next steps (quick):

- Implement Lambda with structured logging and `correlation_id` propagation.
- Add SQS DLQ for failed SNS publishes; enable CloudWatch Alarms on Lambda errors.
- Consider SNS filtering or multiple topics for tenant-specific routing.
- Audit IAM and enable encryption-at-rest for S3 and SNS where required.



Subscription Testing

Validate SNS email confirmations and end-to-end content formatting with sample uploads.

- ❏ Target audience: DevOps and backend engineers. Use this as a deployment checklist and implementation reference.